

JSS2 Mathematics Diagnostic Assessment (Part 2)

Curriculum: Cambridge Lower Secondary (Stages 7 & 8) & Nigerian JSS2 Core
Cluster 2: Algebraic Processes (Questions 21 - 40)

Mathematics - Cluster 2: Algebra

No.	Question & Options	Ans	Diagnostic Focus / Concept Tag
21	Simplify: $4(3x - 5y) - 2(x + 4y)$. A) $10x - 28y$ B) $10x - 12y$ C) $14x - 28y$ D) $10x - 2y$	A	Expansion and simplifying like terms.
22	Solve for x : $3x + 7 = 2(x - 4)$. A) $x = -15$ B) $x = 15$ C) $x = -1$ D) $x = -11$	A	Linear equations with parentheses.
23	Expand: $(x + 4)(x - 3)$. A) $x^2 + x - 12$ B) $x^2 - x - 12$ C) $x^2 + 7x - 12$ D) $x^2 - 12$	A	Quadratic expansion (Cambridge Stage 8).
24	Factorize completely: $6a^2b - 9ab^2$. A) $3ab(2a - 3b)$ B) $ab(6a - 9b)$ C) $3(2a^2b - 3ab^2)$ D) $3a^2b^2(2-3)$	A	Common factor factorization.
25	Find the value of x if $\frac{x}{3} - 4 = 1$. A) $x = 15$ B) $x = 5$ C) $x = 9$ D) $x = 7$	A	Fractional linear equations.
26	Solve the inequality: $2x - 5 > 9$. A) $x > 7$ B) $x < 7$ C) $x > 2$ D) $x > 14$	A	Basic Inequalities.
27	If $y = 2x - 3$, find x when $y = 7$. A) $x = 5$ B) $x = 2$ C) $x = 10$ D) $x = 4$	A	Changing the subject / substitution.
28	Simplify: $\frac{12a^5b^3}{4a^2b}$. A) $3a^3b^2$ B) $8a^3b^2$ C) $3a^7b^4$ D) $3a^{2.5}b^3$	A	Laws of Indices in Algebra.
29	Find the sum of $2p + 3q$ and $4p - 7q$. A) $6p - 4q$ B) $6p + 10q$ C) $2p - 4q$ D) $8p - 21q$	A	Adding algebraic expressions.
30	"Triple a number n and then subtract 5 results in 16." Which equation represents this? A) $3n - 5 = 16$ B) $3(n - 5) = 16$ C) $5 - 3n = 16$ D) $\frac{n}{3} - 5 = 16$	A	Translating word problems to algebra.
31	Expand: $-(2a - 3b)$. A) $-2a + 3b$ B) $-2a - 3b$ C) $2a + 3b$ D) $2a - 3b$	A	Negation and parentheses.
32	Factorize: $x^2 - 25$. A) $(x-5)(x+5)$ B) $(x-5)^2$ C) $(x+5)^2$ D) $x(x-25)$	A	Difference of two squares (JSS3 Readiness).
33	If $a = -2$ and $b = 3$, evaluate $a^2 + 2b$. A) 10 B) 2 C) -2 D) 8	A	Substitution with directed numbers.
34	Solve: $5x + 2 = 3x + 12$. A) $x = 5$ B) $x = 7$ C) $x = 2$ D) $x = 14$	A	Variables on both sides.
35	Simplify: $\frac{x^2}{3} + \frac{x^4}{6}$. A) $\frac{3x^4}{4}$ B) $\frac{x^2}{2}$ C) $\frac{x^6}{6}$ D) $\frac{3x^2}{8}$	A	Algebraic fractions.
36	Find the next term in the sequence: 3, 7, 11, 15, ... A) 19 B) 18 C) 20 D) 21	A	Linear sequences (n^{th} term).
37		A	Reciprocal equations.

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	Solve for p : $\frac{10}{p} = 2$. A) $p = 5$ B) $p = 20$ C) $p = 12$ D) $p = 8$		
38	Which of the following is the same as $a \times a \times a \times b \times b$? A) a^3b^2 B) $3a2b$ C) $(ab)^5$ D) $5ab$	A	Index notation.
39	Solve $4x - 1 \leq 7$. A) $x \leq 2$ B) $x \geq 2$ C) $x \leq 1.5$ D) $x < 2$	A	Linear inequality solution.
40	Diagnostic Insight: Difficulty with expansions (Q23) and factorization (Q32) predicts major struggles in Senior School Mathematics and Physics.	-	Pathway alignment.